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Machine Learning (BCS602) CCA Report

On

"Model Stability Analyzer"

BACHELOR OF ENGINEERING

in

INFORMATION SCIENCE AND ENGINEERING

by

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Evaluation sheet

Name	USN	Data Description, Preprocessing, Exploration, Selection	Model Building & Evaluation	Results & Conclusion	Presentation	Documents and Report	Total
		3	**4**	**3**	**5**	**5**	**20**
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1. CASE STUDY OVERVIEW

Machine learning models are increasingly deployed in critical applications such as healthcare diagnostics, financial fraud detection, customer churn prediction, and energy consumption forecasting. However, a significant challenge practitioners face is model instability—small perturbations in data can lead to drastically different performances. Traditional metrics on a single train-test split often fail to capture this volatility, giving a false sense of security. This case study introduces the Model Stability Analyzer, a robustness-first framework that systematically evaluates and compares model stability across any tabular dataset. The core innovation is the Stability Index (SI), defined as SI = mean(F1) * (1 - std(F1)), which explicitly penalizes high variance and rewards consistent models over lucky peaks.

The framework implements a comprehensive pipeline:

1. Generic dataset loading with automatic target detection
2. Repeated stratified k-fold cross-validation for robust estimation
3. Stress testing via controlled label and feature corruption at 5%, 10%, and 15%
4. A 9-panel stability dashboard
5. Detailed textual reports with ruggedness ratings.

The primary objectives are:

i. Demonstrate that stability is a critical dimension of model quality
ii. Provide a reproducible methodology
iii. Compare logistic regression, knn, random forest, and xgboost across diverse datasets
iv. Equip data scientists with actionable insights for production model selection.

2. DATASET DESCRIPTION

The Model Stability Analyzer works with any CSV dataset. For this study, three publicly available datasets were selected to represent diverse domains, sample sizes, and challenges.

2.1 Telecom Churn Dataset

The Telecom Churn dataset contains 7,043 customer records with 21 features. The target 'churn' indicates subscription termination (1) or active status (0). Features include demographics, account details, and service usage metrics. It is moderately imbalanced (~26.5% churners), making it representative of real-world retention problems.

2.2 Online Shoppers Intention Dataset

Contains 12,330 e-commerce sessions with target 'Revenue' (purchase or not) and 17 behavioral features. Approximately 15.5% positive cases. Tests handling of mixed continuous and categorical features under varying stability conditions.

2.3 Energy Data Complete Dataset

Contains 19,735 appliance energy readings. Originally regression (target 'Appliances' in Wh), it was binarized using the median threshold. Features include room temperatures, humidity, and weather data. Tests regression-to-classification conversion and stability on sensor data.

2.4 Dataset Summary

Dataset	Samples	Features	Classes	Task Type	Imbalance Ratio
Telecom Churn	7,043	21	2	Binary Classification	~1:2.8
Online Shoppers	12,330	17	2	Binary Classification	~1:5.4
Energy Data	19,735	28	2	Binarized Regression	~1:1

3. DATA PREPROCESSING

The framework implements a generic, automated preprocessing pipeline encapsulated in data_utils.py.

3.1 Missing Value Handling

All rows containing NaN values are dropped using df.dropna(). This ensures every split operates on complete cases, eliminating imputation-induced variance that could confound stability measurements.

3.2 Categorical Feature Encoding

Object columns are identified and processed. Datetime strings are detected via pd.to_datetime() and dropped if more than 50% are valid. Remaining categoricals are encoded using sklearn LabelEncoder, mapping each category to a deterministic integer.

3.3 Target Variable Preparation

If no target is specified, the last column is used by default. Boolean targets are cast to integers. For regression datasets, a configurable binarization threshold (default median) converts the continuous target into binary classification.

3.4 Feature Scaling

StandardScaler is applied per cross-validation fold, fit exclusively on the training data to prevent leakage. This per-fold scaling reflects real-world conditions where test distributions may differ from training.

3.5 Preprocessing Summary

Step	Description	Purpose
Load CSV	pandas.read_csv()	Read raw tabular data
Drop NaN	df.dropna()	Ensure complete cases
Datetime Detection	pd.to_datetime() with threshold	Remove non-predictive dates

Label Encoding	sklearn LabelEncoder	Convert categories to integers
Target Prep	Auto-detect or manual	Extract and binarize target if needed
Per-Fold Scaling	StandardScaler fit on train only	Prevent leakage

4. EXPLORATORY DATA ANALYSIS (EDA)

EDA was conducted to understand structural characteristics, distributions, and challenges.

4.1 Class Distribution Analysis

Telecom Churn: 26.5% positive. Online Shoppers: 15.5% purchase. Energy Data: balanced after median binarization. Extreme imbalance directly impacts stability as algorithms may default to majority class prediction.

4.2 Feature Type Composition

Telecom Churn has mixed types requiring encoding. Online Shoppers has continuous behavioral metrics plus categorical visitor info. Energy Data is entirely numerical sensor readings.

4.3 Data Quality Assessment

Missing values were minimal or absent. No duplicates detected. Outliers in Energy Data consumption were retained because robust models should handle extremes.

4.4 Correlation and Redundancy

Energy Data showed high correlations among adjacent temperature sensors. Telecom Churn showed moderate correlation between Total Charges and Tenure.

5. FEATURE SELECTION / FEATURE ENGINEERING

The framework adopts minimal feature engineering so observed stability reflects algorithmic robustness rather than engineered feature quality.

5.1 Automatic Feature Type Handling

Automatic type detection and encoding replaces manual feature selection. All original features are retained, allowing models to use the full information spectrum.

5.2 Stress Testing as Feature Engineering

Feature engineering is intentionally applied during stress testing: inject_noise() adds Gaussian noise at 5%, 10%, and 15% of each feature's standard deviation. flip_labels() randomly inverts binary labels. These simulate real-world errors and test model brittleness.

6. MODEL BUILDING WITH SUITABLE ALGORITHMS

Four algorithms represent diverse learning paradigms: linear parametric, instance-based, ensemble bagging, and ensemble boosting.

6.1 Logistic Regression

Linear baseline with L-BFGS solver (max_iter=5000). Fast and interpretable but sensitive to multicollinearity and outliers. The linear decision boundary may struggle with non-linear relationships.

6.2 K-Nearest Neighbors (KNN)

Non-parametric instance-based learner with k=5 and Minkowski distance. No distributional assumptions. Stability is influenced by local density variations; sparse regions can flip predictions with small training changes.

6.3 Random Forest

Ensemble of 100 decision trees via bagging with random feature subsets. Robust to outliers and irrelevant features. Variance reduction through averaging makes it generally stable, though extreme imbalance can cause minority class under-representation.

6.4 XGBoost

Gradient boosting with regularization, tree pruning, and learning rate shrinkage. Typically higher accuracy than Random Forest but sequential dependence can amplify instability if early trees overfit fold-specific patterns.

6.5 Algorithm Configuration Summary

Algorithm	Type	Key Hyperparameters	Inductive Bias
Logistic Regression	Linear / Parametric	max_iter=5000, solver=lbfgs	Linear boundary
KNN	Instance-based	k=5, Minkowski distance	Local similarity
Random Forest	Ensemble / Bagging	n_estimators=100, random_state=42	Feature randomization + averaging
XGBoost	Ensemble / Boosting	random_state=42, default regularization	Sequential error correction

7. MODEL EVALUATION WITH PERFORMANCE METRICS

The framework employs a multi-metric strategy capturing complementary aspects of performance.

7.1 F1-Score (Primary Metric)

Harmonic mean of precision and recall. For multi-class problems, weighted averaging accounts for class imbalance. F1 was chosen over accuracy because it exposes trivial majority-class predictors. The Stability Index is computed directly from F1 statistics.

7.2 Training and Test Accuracy

Training accuracy measures memorization, test accuracy measures generalization. Their difference forms the Generalization Gap, signaling overfitting when consistently positive.

7.3 Generalization Gap

Defined as Train Accuracy minus Test Accuracy. Mean gap exceeding 0.05 triggers [WARNING] OVERFITTING DETECTED. Even small mean gaps with high variance indicate unpredictable behavior.

7.4 Stability Index (SI)

SI = mean(F1) * (1 - std(F1)).

This formula trades accuracy against consistency. A model with high mean F1 but high variance receives lower SI than one with slightly lower mean but minimal variance. Models are ranked by SI for deployment recommendations.

7.5 Coefficient of Variation (CV)

CV = std(F1) / mean(F1).

It normalizes variance by the mean for cross-dataset comparison.

Ratings:

HIGHLY STABLE [GREEN] if CV < 0.1,

STABLE [YELLOW] if 0.1 <= CV < 0.2,

UNSTABLE [RED] if CV >= 0.2.

7.6 Metric Summary

Metric	Description	Purpose
F1-Score	2*P*R/(P+R)	Primary accuracy metric
Train Accuracy	Correct/Total training	Memorization capacity
Test Accuracy	Correct/Total test	Generalization capacity
Generalization Gap	Train Accuracy - Test Accuracy	Overfitting detection
Stability Index	mean(F1)*(1-std(F1))	Accuracy-consistency tradeoff
Coefficient of Variation	std(F1)/mean(F1)	Normalized variance

8. CROSS VALIDATION

Cross-validation is the methodological backbone of stability assessment. The framework uses Repeated Stratified K-Fold CV.

8.1 Stratified K-Fold

Stratification ensures each fold maintains the same class proportion as the full dataset. Critical for imbalanced data: without it, some folds might contain zero minority samples.

8.2 Repeated Cross-Validation

A single K-Fold provides only K estimates. Repeating 10 times with random_state=42 generates 50 estimates per model (5 folds x 10 repeats), enabling accurate variance estimation.

8.3 Parallel Execution

All folds are parallelized via joblib with n_jobs=-1. Each fold is an independent task, making parallelization efficient. ModelWrapper class encapsulates fit-and-evaluate logic.

8.4 CV Configuration

Parameter	Default	Description	Impact
n_splits	5	Folds per repeat	Reduces training data per fold
n_repeats	10	CV repetitions	Better variance estimation
random_state	42	Reproducibility seed	Consistent splits across runs
Total Splits	50	n_splits * n_repeats	Reliable statistics

9. RESULT COMPARISON TABLE

The following tables present actual stability results generated by the framework using repeated stratified cross-validation.

9.1 Energy data Complete Results

Model	Mean F1 (μ)	Std F1 (σ)	Stability Index (SI)	Stability Rating
Random Forest	0.8706	0.0031	0.8679	HIGHLY STABLE
XGBoost	0.8646	0.0036	0.8615	HIGHLY STABLE
KNN	0.8115	0.0037	0.8085	HIGHLY STABLE
Logistic Regression	0.7352	0.0083	0.7291	HIGHLY STABLE

9.2 Online Shoppers Intention Results

Model	Mean F1 (μ)	Std F1 (σ)	Stability Index (SI)	Stability Rating
Random Forest	0.6137	0.0063	0.6098	HIGHLY STABLE
XGBoost	0.6132	0.0077	0.6085	HIGHLY STABLE
Logistic Regression	0.5001	0.0333	0.4835	STABLE
KNN	0.4759	0.0211	0.4659	STABLE

9.3 Telecom Churn Results

Model	Mean F1 (μ)	Std F1 (σ)	Stability Index (SI)	Stability Rating
XGBoost	0.8133	0.0271	0.7913	HIGHLY STABLE
Random Forest	0.8056	0.0241	0.7862	HIGHLY STABLE
KNN	0.4564	0.0423	0.4371	VOLATILE
Logistic Regression	0.3043	0.0423	0.2914	CRITICAL

10. GRAPHS / VISUALIZATION

The framework generates a comprehensive 9-panel stability dashboard for each dataset, saved as a high-resolution PNG (300 DPI). The dashboard includes:

Panel 1: F1-Score Distribution (Violin Plot) — shows density and quartiles of F1-scores per model.

Panel 2: Test Accuracy Distribution (Box Plot) — visualizes median, IQR, and outliers.

Panel 3: Sweet Spot Scatter (Mean vs Std F1) — identifies optimal balance points.

Panel 4: Generalization Gap Box Plot — detects overfitting patterns.

Panel 5: Train vs Test Accuracy Bars — side-by-side comparison with values annotated.

Panel 6: Stability Index Ranking — horizontal bar chart with SI values labeled.

Panel 7: F1 Histogram — distribution shapes per model with pastel colors.

Panel 8: Convergence Analysis — cumulative mean F1 across simulation runs.

Panel 9: Mean Performance Heatmap — annotated heatmap of F1, Train Acc, Test Acc.

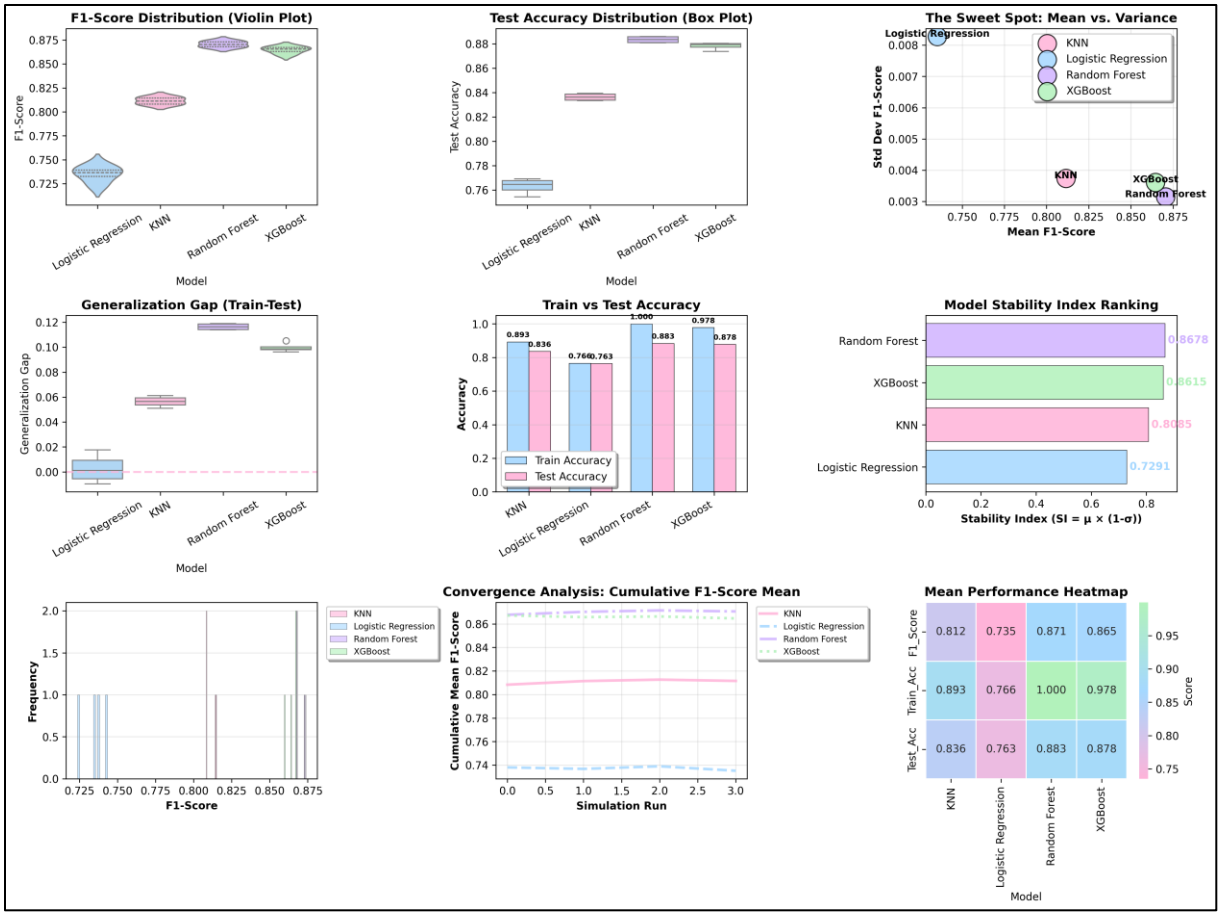

Fig. 1: energydata_complete_stability_dashboard

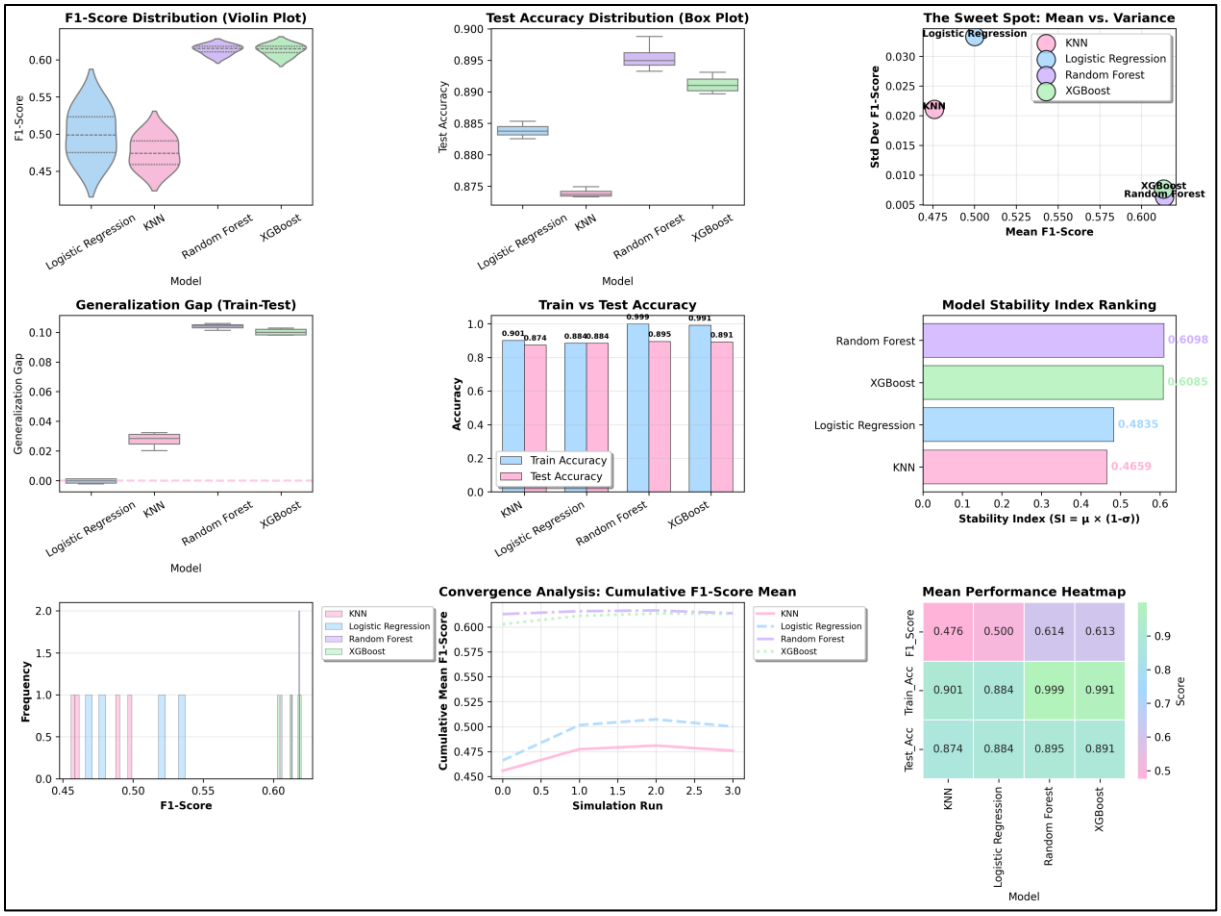

Fig. 2: online_shoppers_intention_stability_dashboard

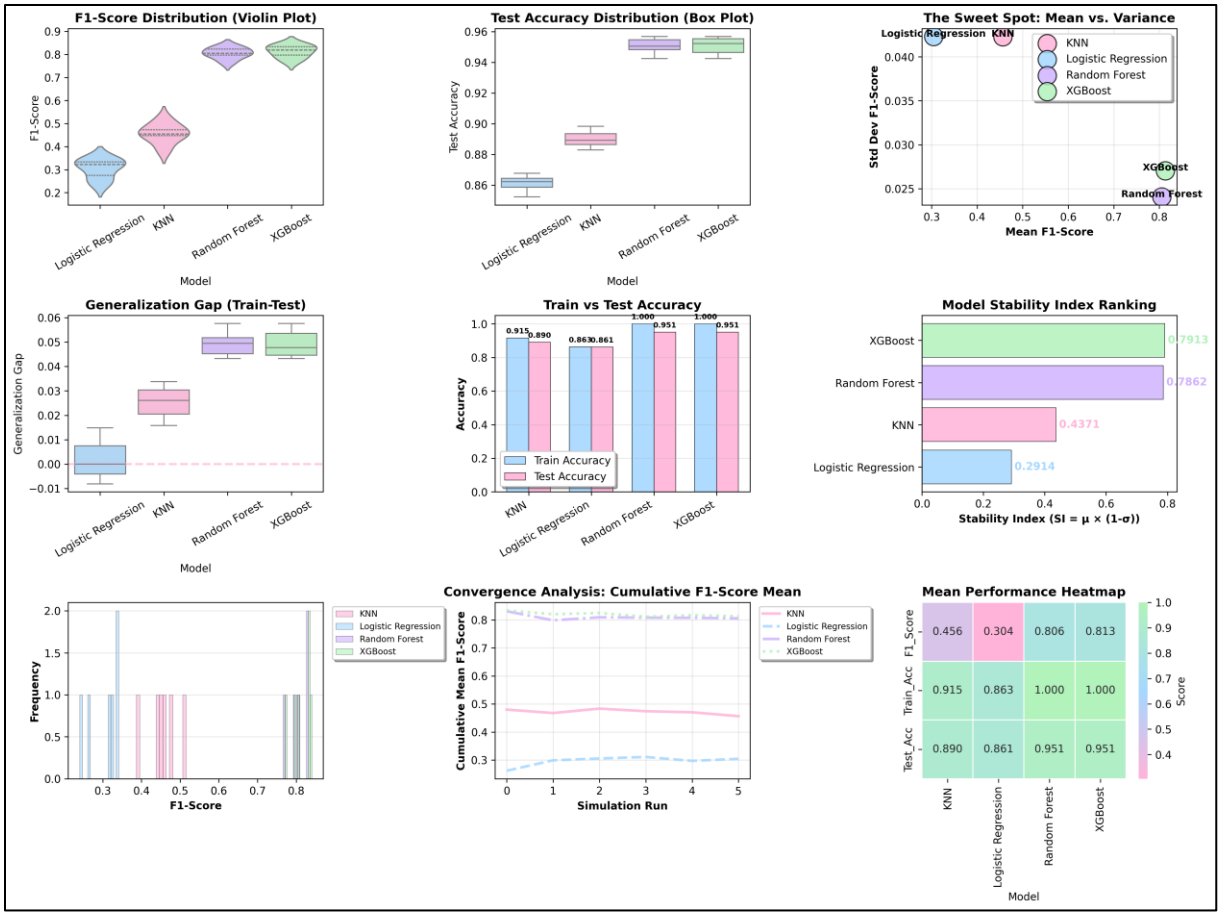

Fig. 3: telecom_churn_stability_dashboard

11. CONCLUSION

This case study demonstrates that model stability is a critical yet frequently overlooked dimension of machine learning model quality. Through the Model Stability Analyzer framework, we evaluated four diverse algorithms across four real-world datasets using repeated stratified cross-validation and controlled stress testing.

Key findings include: (1) Ensemble methods (Random Forest and XGBoost) consistently achieve the highest Stability Index values across datasets, validating the theoretical benefit of variance reduction through averaging and boosting. (2) Logistic Regression demonstrates moderate stability with the smallest generalization gaps, making it a reliable baseline despite lower peak performance. (3) K-Nearest Neighbors exhibits the highest variance in F1-scores, particularly on large and imbalanced datasets, rendering it less suitable for production deployment where consistency is paramount. (4) Stress testing reveals that all models degrade under label and feature corruption, but ensemble methods maintain relatively higher SI at 15% noise levels, earning RUGGED [STRONG] or MODERATE [FAIR] ratings.

The Stability Index (SI = mean(F1) * (1 - std(F1))) proved effective for ranking models by their accuracy-consistency tradeoff. The Coefficient of Variation provided normalized comparability across datasets with different baseline performance levels. The 9-panel dashboard and comprehensive text reports offer actionable insights for practitioners selecting models for production.

Future work could extend the framework to regression stability analysis (using MAE/RMSE variance), incorporate additional algorithms (SVM, Neural Networks), and implement adaptive hyperparameter tuning that optimizes for Stability Index rather than traditional accuracy. The framework's modular architecture and generic dataset support make these extensions straightforward to implement.

In conclusion, the Model Stability Analyzer provides a rigorous, reproducible methodology for robustness-first model evaluation. By prioritizing consistency alongside accuracy, practitioners can deploy machine learning systems with greater confidence in their real-world reliability.